PEMS — Picking Exception Management System

Objective: Build a PostgreSQL- and Python-based decision-support prototype that identifies recurring picking exceptions at warehouse storage locations, prioritises suspicious locations, and generates investigation recommendations for review by a captain.

Primary problem: Incorrect product placement and recurring product-location mismatches. A single rejected scan is not sufficient evidence of a shelf problem. Repeated mismatches involving the same location, potentially the same unexpected product, and multiple pickers provide stronger evidence for physical investigation.

Secondary problems: Damaged or unreadable barcodes, missing stock, and mixed physical products. These must be represented separately. Mixed-item errors are not directly detectable from the currently described scan data alone.

Solution: Analyse picking events and scan attempts, identify recurring location-level patterns, calculate an explainable risk score, and produce suggested quality tasks. A captain reviews recommendations and may assign a quality shopper. Physical inspection outcomes are stored separately and used to evaluate detection accuracy.

Data: PostgreSQL database picking_exceptions, Python, psycopg2, and a reproducible synthetic dataset of 5,000 picking events, approximately 1,500–2,000 orders, 100 products, 110 barcodes, 40 locations, 20 pickers, five quality shoppers, and two captains.

Simulation: Normal picking, damaged barcodes, incorrect shelf products, mixed physical products, and missing stock. Five recurring hotspot locations with repeated mismatch patterns across different pickers. Fixed random seed and timezone-aware timestamps. Expected products and scanned products must remain distinct.

Ground truth: Store scenario labels in a separate evaluation-only table. Never expose those labels to the detection pipeline. Evaluate precision, recall, false-positive rate, detection delay, and investigation outcomes.

Implementation: Finalise schema; generate synthetic data; conduct SQL analysis; develop baseline detection; implement risk scoring; generate suggested tasks; record investigation outcomes; evaluate; then investigate mixed-item verification as a second module.

Key principle: Distinguish an observed exception, a suspected recurring operational problem, and a physically confirmed cause. Do not treat every rejected scan as incorrect shelf placement or claim to detect what the available data cannot reveal.

Expected deliverable: A reproducible, documented end-to-end prototype converting warehouse picking-event data into explainable, prioritised quality-investigation recommendations.